INTEGRITY PROTOCOL

Comprehensive Design & Specification

Version 0.3 (Consolidated)

Foundational Primitives · AIS · Memory · Oracle · Interop

Normative intent for the protocol, with implementation map. Package coordination lives in `docs/INTERFACE_CONTRACT.md`; wire surfaces under `spec/`.

> ***Ground rule:*** *No silent mocks. Every claim is implemented and tested against a real toolchain, or explicitly marked [PLANNED] / [PARTIAL].*

Xibalba Tech Solutions

Agentic Trust Layer for the AI Economy

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1. Purpose and Thesis

Integrity Protocol is a **trust and compliance layer for the agentic economy**. It answers two questions that pure off-chain systems cannot answer honestly:

1. **Regulatory compliance** — Can a regulator or counterparty verify an AI agent's behavior *after the fact*, without trusting the agent's own word?
2. **Agent trust** — Can one agent (or service) verify another's track record *before* transacting with it?

Thesis: An agent must be an **Economic Sovereign** — a continuing entity that owns its smart contracts, remembers via durable anchored memory, can bind future behavior, holds material value at risk, produces independently checkable evidence, and cannot unilaterally rewrite that evidence after finality.

AIS (Agent Integrity Score) is the **credit-score layer**. The primitives below are the **institutional facts** that make the score meaningful.

2. Origin: Credit-Score Analogy

The project began by defining metrics analogous to a human credit score. A credit score only works because deeper institutional facts exist:

Human substrate	Agent analogue
Persistent legal identity	Cryptographic self-sovereignty + persistent memory
Exclusive control of accounts	Agent-owned smart contracts
Binding commitments	Behavioral Commitment (BCC)
Loss on default	Bonded stake
Observable history	Continuous verifiable observability
Hard-to-rewrite records	Immutable on-chain anchors
Composite score	AIS

AIS is downstream of primitives — not a replacement for them.

3. Foundational Properties of the Medium

3.1 Immutability of Anchored History

Once a commitment is accepted by the chain and finalized, it cannot be altered or deleted by the agent, the operator, or any single party.

Without immutability	With immutability
Scores, roots, registrations can be rewritten	Deployments, roots, stake, score updates are fixed facts
Audit trusts the current DB owner	Audit checks inclusion against finalized chain state

Normative rules:

1. Identity-establishing transactions (deploy SovereignAgent / StateAnchor, registerPrimitives, genesis memory root) are on-chain.
2. Trust-affecting transitions (score updates, disputes, slashes) MUST be on-chain or checkpointed on-chain.
3. Oracle DB and dashboards are **caches**; in conflict, **finalized chain wins**.

4. Mutable off-chain records MUST NOT be presented as equivalent to anchored history.

Immutability means **append-only, prior commitments remain verifiable** — not “nothing changes.” StateAnchor never un-anchors an old root (isAnchoredRoot stays true).

3.2 Public Verifiability

Anyone with chain access can re-check deployments, roots, stake, and score updates without permission from the agent or operator.

3.3 Attribution via Signatures

State transitions are authorized by cryptographic keys bound to agent-owned contracts — a clear answer to “who caused this history?”

4. Foundational Primitives

Derived from economic principles (continuity, residual control, exclusive identity rights, credible commitment, internalization of consequences, verifiability); enforced by the medium properties above.

4.1 Persistent Memory

Principle: Continuity of the economic agent. The agent MUST control a durable Trust Vault whose commitments can be Merkle-anchored on its StateAnchor. Registration requires StateAnchor.latestRoot ≠ bytes32(0). The genesis root (epoch 0→1) MUST be agent-authorized (controller or SovereignAgent.execute). Later roots MAY use protocol ANCHOR_ROLE. Empty-but-initialized vault is valid at birth. Raw vault stays agent-controlled. **Status:** Architecturally supported; registration gate enforcement [PLANNED].

4.2 Agent-Owned Smart Contracts

Principle: Residual rights of control. The agent’s own key deploys SovereignAgent + StateAnchor; five EIP-1167 clones take admin = SovereignAgent address. Post-registration changes route through execute(...). Deployment transactions are immutable proof of self-sovereign creation. **Status:** Implemented.

4.3 Cryptographic Self-Sovereignty

Principle: Exclusive property rights in identity. did:integrity:<sha256(pubkey)> (Ed25519) + secp256k1 controller; rotation without erasing history; protocol never custodies keys. **Status:** Implemented.

4.4 Behavioral Commitment (BCC)

Principle: Credible commitment. Signed pre-execution intent (identity, intent type, state hash, nonce, timestamp, optional covered entity). Required for regulated/high-value actions. Outcomes SHOULD become vault leaves. **Status:** Implemented (SDK, CLI, bcc_middleware + OPA).

4.5 Economic Skin-in-the-Game (Bonded Stake)

Principle: Internalization of consequences. Collateral in per-agent Slasher; agent cannot arbitrate own disputes; lock/slash history on-chain. **Status:** Contracts + dispute path exist; uniform minimum stake at registration [PARTIAL].

4.6 Continuous Verifiable Observability

Principle: Monitoring for reputation and regulation. Signed, redacted telemetry; oracle **re-derives** signals; silence while acting is observable [PLANNED as automated signal]. Counterparties rely on proofs against finalized roots, not the oracle's private DB alone. **Status:** Ingest, re-derivation, SSE implemented.

5. On-Chain Architecture

Every registered agent owns **seven primitives**:

#	Contract	Deploy	Role
1	SovereignAgent	Direct	Identity, DID, execute, controller rotation
2	StateAnchor	Direct	Merkle roots of Trust Vault
3	ReputationRegistry	Clone	AIS ledger + ZK-boost bookkeeping
4	Slasher	Clone	Bonded stake; dispute-gated slashing
5	VerifierRegistry	Clone	Versioned ZK verifier pointer
6	ComplianceGate	Clone	Vertical declaration + live check
7	AgentProfile	Clone	Domain membership + metadata

XibalbaAgentRegistry indexes each agent's PrimitiveSet. Downstream contracts resolve **per-agent clones**, not global singletons. Call-routing: clone admin = SovereignAgent contract. Bootstrap exception: registerPrimitives is EOA-signed once. Merkle: keccak256 leaves; sorted-pair parents.

6. Registration

Economic birth of the agent. Required sequence (normative intent):

- Create/load DID + controller wallet; bind them.
- Fund wallet; prepare stake.
- Deploy SovereignAgent + StateAnchor under agent key.
- Agent-authorize **genesis memory root** on StateAnchor.
- AgentPrimitivesFactory.registerPrimitives (atomic PrimitiveSet).
- Meet minimum bonded stake [PARTIAL].
- Oracle re-verifies on-chain primitives **and** non-zero genesis root; then accepts agent.

No half-registered agents. **Current repo gap:** genesis root and memory check are not yet enforced.

7. Persistent Memory, Genesis Root, Lineage

7.1 Oracle Enforcement (Target)

After PrimitiveSet match, oracle reads StateAnchor.latestRoot. If zero → 400 MemoryNotInitialized. Same independent-read posture as primitive re-verification.

7.2 Genesis Root Authorization

First root MUST come from controller / SovereignAgent.execute (e.g. one-time initializeMemory when latestEpoch == 0). Protocol ANCHOR_ROLE for epoch ≥ 2 only.

7.3 Copying Another Agent's Memories

Action	Effect
Copy vault bytes	Does not transfer identity, stake, or AIS
Claim another's roots/AIS	Reject — bound to original StateAnchor / registries
Steal keys + memory	Ordinary account takeover; recovery via rotation if available

7.4 Lineage (Fork / Migration / Recovery)

Explicit controller-signed attestation + on-chain record. Default: no automatic AIS/stake transfer. Optional capped partial credit after challenge window under reputation_policy: partial. New agent still performs its own genesis root; may include lineage hash as a vault leaf.

7.5 Behavioral Similarity

Observational only in current version. MUST NOT alone cause registration rejection or automatic slash. MAY later inform soft AIS ceilings or dispute evidence under versioned rules.

8. Agent Integrity Score (AIS)

8.1 Formula

```
AIS = (S_entropy·wE + S_grounding·wG + S_sacrifice·wS + S_compliance·wC) · ZK_boost
```

Default weights: 0.30 / 0.30 / 0.20 / 0.20 (sum 1.0). ZK_boost = 1.15 if any verified Barretenberg proof in period; else 1.0. Each S_* ∈ [0, 1000]. Final AIS is **not** clamped (boost can exceed 1000). Sole computer: integrity-oracle/scoring-core. All consumers use GET /v1/agent/{id}/ais.

8.2 Component Intuition

Component	Credit-score parallel	Implementation input
Entropy	Stability	performance_variance (lower better)
Grounding	Evidence-based action	hgi_raw ∈ [0,1]
Sacrifice	Costly effort	hours-equivalent token proxy
Compliance	Clean vs flagged	penalty_ratio
ZK_boost	Costly cryptographic evidence	bb verify in window

8.3 Trust Model

Signature proves **who**. Oracle **re-derives** entropy/grounding/sacrifice from signed span content; compliance mixes flags + live ComplianceGate. Client derived_signals are audit trail only. Identity ceiling [PLANNED]: AIS_final = min(AIS, Tier_ceiling).

9. Oracle Implementation

9.1 Pipeline

Signed POST /v1/telemetry/ingest → verify sig + nonce (FOR UPDATE) + PHI scan → derive::recompute(otel_spans) → compliance (self-report; chain wins if covered_entity set) → optional bb verify → insert telemetry_events → aggregate_for_ais (~30 days) → AisEngine::score → GET /ais + SSE AisUpdate →

bcc_middleware pushes updateScore / raiseDispute.

9.2 scoring-core Maps

- Entropy: $\exp(-1.5 \cdot v^2) \cdot 1000$
- Grounding: hgi·1000
- Sacrifice: min(log10(hours+1)/3, 1)·1000 (saturates ~1000h)
- Compliance: (1–penalty)·1000

9.3 derive.rs (Authoritative Inputs)

Lexical stability = inverted normalized Shannon entropy over words. Grounding = keyword heuristic (uncertainty markers → 0.40 else 0.95). Sacrifice = Σtokens / 50,000 hours proxy. Store variance as 1.0 – stability for score polarity.

9.4 Known Limitations

- ZK boost is period-wide BOOL_OR, not per-event bound.
- Grounding is keyword-only; sacrifice is token proxy, not verified GPU.
- covered_entity_address is client-supplied (spoof residual).
- Tier ceiling not applied in formula yet.
- Oracle global Merkle helpers largely unused; real anchors via bcc_middleware per-agent.

10. Telemetry and OpenTelemetry

Integrity has **two parallel span paths**. Confusing them is a primary source of integration bugs.

Path	Transport	Auth	Storage	AIS?
A. Signed ingest	HTTP /v1/telemetry/ingest	Signature	telemetry_events	Yes
B. Real OTLP	gRPC :4317	None (rate limit)	otel_spans	No

JSON field name otel_spans on path A ≠ OTLP table otel_spans on path B. Path A is a flat tagged JSON array (kind: telemetry | trace_run | custom_metrics) optimized for scoring. Path B requires resource attribute integrity.agent.id; used for trace trees and UI only.

Standard OTel span fields: name, trace_id, span_id, parent, kind, start/end, attributes, events, links, status — under ResourceSpans → ScopeSpans → Span.

11. Behavioral Commitment (BCC)

Pre-execution signed object: agent_id, intent_type, intended_state_hash, nonce, timestamp, agent_public_key, optional covered_entity_address, signature. Canonical JSON: sorted keys, ensure_ascii=True, compact separators — shared across SDK, CLI, middleware. OPA + on-chain BAA checks for healthcare intents. Fail closed when BAA required but unverifiable.

12. Verification Ladder

Tier	Verification	AIS ceiling	Status
1 Sovereign	Software key	600	Default; ceiling not enforced

Tier	Verification	AIS ceiling	Status
2 Linked	DNS / social	850	Not built
3 Institutional	TEE + audit	1000	Verifier exists; unwired

Higher tier raises ceiling; does not replace primitives.

13. Xibalba Shield

HIPAA/healthcare flagship: Smart BAAs, ComplianceGate, EHRGate, redaction before egress, immutable audit roots. Proves primitives under maximum regulatory pressure. Not a side feature — the credibility proof for the protocol.

14. Proprietary LLMs and Attribution

14.1 Proprietary Models

Allowed as **tools behind the agent boundary**. Protocol does not require open weights. Limits depth of cognitive verification; does not block the Economic Sovereign shell. Must: commit-then-call (BCC); log model id/version; redact before vendor egress; claim only agent-boundary verifiability.

14.2 Novelty vs Attribution

Integrity does **not** score literary novelty. It scores **attributable, continuous, staked behavior**. Convergence on professional standards is expected (not penalized). Parasitic cloning of another agent's anchored history confers no reputation. Prompts/RAG/vault create private advantage; the protocol prevents laundering that into another identity's score — it is not a copyright office.

15. On-Chain Reputation and ERC-8004

Landscape families include personhood (World ID, BrightID), stamp aggregation (Gitcoin Passport), subjective feedback, stake/slash systems, contribution graphs, and agent registries (**ERC-8004: Trustless Agents**).

ERC-8004 defines three registries: Identity (ERC-721 handle), Reputation (client feedback), Validation (stake/zk/TEE/judges). It is discovery + pluggable signals — not a full Economic Sovereign stack.

Concern	System of record
Find the agent	ERC-8004 Identity
Trust with value / PHI	Integrity primitives + AIS
Optional market feedback	ERC-8004 Reputation / Validation

Integrity DID + SovereignAgent remain canonical. 8004 NFT is a discovery pointer; transfer must not silently move Integrity reputation. Optional: publish primitive addresses and AIS endpoint in agentURI.

16. Package Map and Status

Package	Role	Status
contracts/	7 primitives, factory, Shield, token, verifiers	Mature; Base Sepolia

Package	Role	Status
integrity-zkp/	Noir + Barretenberg intent circuit	Real pipeline
integrity-oracle/	Telemetry, AIS, chain reads, OTLP, SSE	Solid + e2e
integrity-sdk/ / cli	DID, wallet, registration, BCC, telemetry	Solid
bcc_middleware/	Pre-exec gate, OPA, anchor, score sync	Solid
integrity-userapi/	Off-chain users / API keys	In progress
integrity-mvp/	Product surface + demo engine	In progress

17. What This Protocol Is Not

- Not an agent runtime or LLM router
- Not a custodial key service
- Not a claim that off-chain databases are immutable
- Not “AIS alone = trust” without primitives
- Not end-to-end audit of proprietary model weights
- Not a novelty or copyright engine
- Not a guarantee of honest behavior — only that behavior is costly to fake, verifiable, and bound to a continuing economic subject whose commitments cannot be unilaterally rewritten

18. Document Relationship and Revision Policy

Document	Role
This specification	Foundational properties, primitives, AIS, memory, interop
docs/INTERFACE_CONTRACT.md	Internal schemas, ports, sequences, env
spec/*	External versioned wire surfaces
PRODUCTION_GAPS.md	Honest open items
docs/wiki/	Concept and entity pages

Revision: Additive clarification in 0.x is allowed. Removing or weakening a foundational primitive, or treating mutable off-chain state as equivalent to finalized anchors, requires a major version and migration plan. Unimplemented norms stay marked [PLANNED] / [PARTIAL].

Appendix A — Priority Implementation Gaps

1. Enforce genesis memory root + oracle latestRoot ≠ 0 at registration.
2. Agent-authorized initializeMemory (or equivalent) on StateAnchor.
3. Uniform minimum stake at registration / tier elevation.
4. Tighter ZK-boost binding (per-event / public inputs).
5. Identity-ceiling clamp in scoring.
6. Lineage attestation + on-chain record.
7. Optional ERC-8004 registration adapter for discovery.
8. Silence-as-signal for observability obligation.

Appendix B — One-Sentence Summary

Integrity Protocol makes AI agents Economic Sovereigns: they own their contracts, anchor their memory on an immutable ledger, commit before high-stakes action, stake capital against misbehavior, and emit independently re-derived evidence — so counterparties and regulators can trust a score that rests on facts, not self-report.

End of specification (v0.3)

Xibalba Tech Solutions · Integrity Protocol